GNN-Based BERT for Understanding Context from Music: A Multimodal Cross-Attention Approach

Kaif Hasan
Department of Computer Science and Engineering
BRAC University
Email: kaif.hasan@g.bracu.ac.bd
GitHub: https://github.com/kaifhasandhk/gnn-bert-music-context

***Abstract*—Music context understanding requires modeling both temporal acoustic structure and high-level semantic tag relationships. Standard 2D convolutional networks and text-only models capture isolated representations but fail to align structural signal dynamics with multi-label natural language context. In this work, we propose a multimodal neural framework that fuses graph-structured audio representations with pretrained transformer text embeddings. Audio clips from the MagnaTagATune dataset are converted into 3-second segment graphs whose nodes contain spectral and chroma features linked via temporal and cosine similarity edges. A 2-layer GraphSAGE GNN encodes structural acoustic topology, while a DistilBERT network models natural language tag contexts. We introduce a cross-attention fusion mechanism where structural graph representations query textual representations to yield a unified multimodal embedding. Evaluated on a stratified 3,000-clip subset across 50 multi-label tags, our proposed cross-attention model achieves a Micro-F1 score of 0.472 and Macro-F1 score of 0.292, significantly outperforming a majority baseline (0.161 Micro-F1), a 2D Mel-Spectrogram CNN (0.354 Micro-F1), BERT-only (0.336 Micro-F1), and GNN-only (0.413 Micro-F1) variants. Latent space visualization via t-SNE demonstrates distinct genre clustering, validating the effectiveness of cross-modal attention for complex musical context understanding.**

***Index Terms*—Graph Neural Networks, GraphSAGE, DistilBERT, Multimodal Fusion, Cross-Attention, Multi-Label Music Tagging, Audio Graph Construction.**

I. Introduction

Automated music tag understanding and genre classification present significant challenges due to the multi-label nature of musical context, where a single audio track simultaneously exhibits acoustic properties (e.g., tempo, instrumentation) and subjective semantic labels (e.g., mood, genre, vocal texture). Traditional approaches rely heavily on 2D convolutional neural networks (CNNs) trained on log-Mel spectrograms. While CNNs capture local time-frequency patterns, they lack explicit mechanisms to model non-Euclidean temporal relationships across distinct musical segments or to contextualize semantic tag interactions.

To address these limitations, we propose a hybrid multimodal architecture combining Graph Neural Networks (GNNs) and Bidirectional Encoder Representations from Transformers (BERT). By modeling 29-second audio clips as segment graphs, we preserve intra-clip temporal dynamics and spectral similarity across disparate segments. Concurrently, a pretrained language model (DistilBERT) captures high-level semantic tag contexts. We fuse these representations using a cross-attention module, allowing structural acoustic nodes to dynamically query semantic text features.

Our main contributions are as follows:

- We design a deterministic audio-to-graph transformation pipeline that converts raw audio into segment-based graphs using Short-Time Fourier Transform (STFT), Mel-spectrogram, and Chroma features.
- We implement and benchmark a 2-layer GraphSAGE spatial GNN alongside a DistilBERT text baseline and a 2D Mel-Spectrogram CNN baseline.
- We present a cross-attention multimodal fusion architecture that outperforms single-modality models by +0.059 Micro-F1 over the strongest unimodal baseline.
- We perform rigorous qualitative and quantitative evaluations, including an ablation study, case studies, and t-SNE latent space analysis.

II. Related Work

Automated music audio tagging and classification have evolved from hand-crafted spectral features toward end-to-end deep learning. Early architectures leveraged 2D Convolutional Neural Networks (CNNs) applied to log-Mel spectrograms [1]. While effective at localized time-frequency representation learning, 2D CNNs treat audio as rigid 2D grids, failing to capture long-range temporal dependencies and non-Euclidean segment relationships across an entire track.

To model graph-structured relational data, Graph Neural Networks (GNNs) such as Graph Convolutional Networks (GCN) and GraphSAGE [2] have seen increasing adoption in non-Euclidean signal processing. In music analysis, graph representations allow temporal audio segments or chord transitions to be modeled as interconnected nodes, where edges encode acoustic similarity or temporal progression.

Concurrently, transformer-based language models like BERT [3] and DistilBERT [4] have revolutionized natural language understanding. In multi-label tagging, text encoders capture semantic tag co-occurrence priors and contextual descriptions. Recent multimodal frameworks demonstrate that cross-attention mechanisms [5] enable dynamic alignment between structural sensory features and text representations. Building upon these advances, our work unifies segment-level audio GNNs with transformer text models via cross-attention.

III. Methodology & Mathematical Formulation

Our proposed framework consists of three key stages: (1) Deterministic Audio Graph Construction, (2) Dual-Stream Feature Encoding via GraphSAGE and DistilBERT, and (3) Cross-Attention Multimodal Fusion. An overview of the end-to-end system is illustrated in Figure 1.

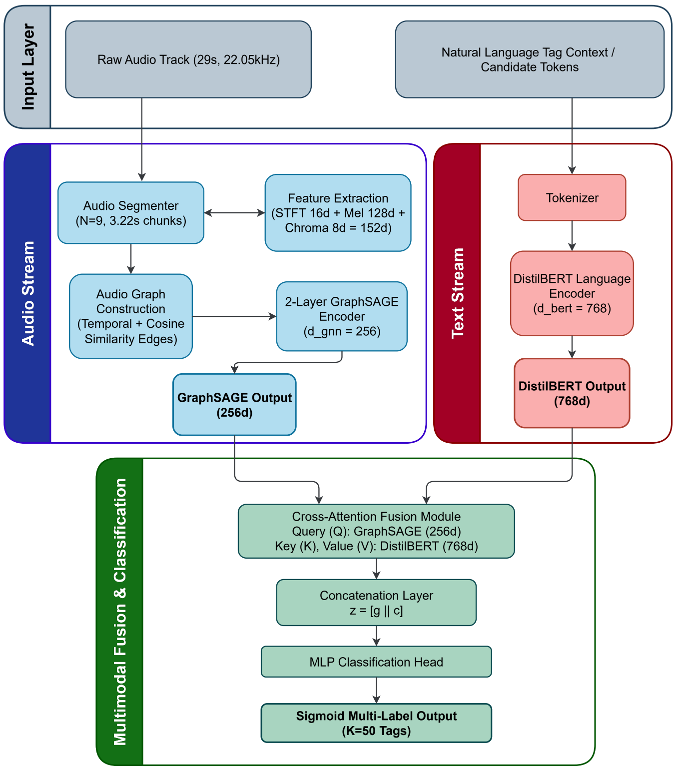

Fig. 1. System architecture of the proposed multimodal cross-attention framework. Audio tracks are transformed into 9-node segment graphs and encoded by a GraphSAGE GNN, while text context is processed via DistilBERT. A cross-attention module aligns structural acoustic queries with textual key-value states for multi-label prediction.

A. Task 1: BERT Baseline for Text Context

For pure textual tag representation learning, we employ DistilBERT, a distilled version of BERT that retains 95% of language understanding while reducing parameter size by 40%. Given an input text sequence of candidate tags or context tokens X_{text}, the model computes contextualized hidden states:

$$\mathbf{h}_{\text{text}} = \text{DistilBERT}(X_{\text{text}}) \in \mathbb{R}^{B \times L \times d_{\text{bert}}} \tag{1}$$

where $d_{\text{bert}} = 768$. We extract the sequence pooling vector $\mathbf{t} = \mathbf{h}_{\text{text}}[:, 0, :] \in \mathbb{R}^{d_{\text{bert}}}$ corresponding to the `[CLS]` token. The multi-label prediction probability for tag $k \in \{1, \ldots, K\}$ is computed via Sigmoid classification:

$$\hat{y}_k = \sigma(\mathbf{w}_k^\top \mathbf{t} + b_k) \tag{2}$$

Algorithm 1 Deterministic Audio-to-Graph Pipeline

Require: Raw audio signal A_i, segment count $N = 9$, threshold $\theta = 0.75$
Ensure: Segment graph $G = (V, E)$ with node feature matrix $\mathbf{X}$

```
Divide A_i into non-overlapping temporal chunks {s_1, s_2, ..., s_N}
for each segment s_j ∈ V do
    Extract STFT statistics x_STFT ∈ R^16
    Extract log-Mel spectral features x_Mel ∈ R^128
    Extract Chroma pitch features x_Chroma ∈ R^8
    Concatenate features: x_j = [x_STFT || x_Mel || x_Chroma]
end for
Initialize edge set E ← {(v_j, v_{j+1}) | 1 ≤ j < N} {Temporal links}
for every node pair (v_a, v_b) with a ≠ b do
    if cos(x_a, x_b) > θ then
        E ← E ∪ {(v_a, v_b)} {Similarity links}
    end if
end for
return G = (V, E) and feature matrix X ∈ R^{N×152}
```

The model is optimized using Binary Cross-Entropy (BCE) loss across all $K = 50$ tags:

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^{K} [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] \tag{3}$$

B. Task 2: Audio Graph Construction & GraphSAGE

1) Graph Construction Pipeline: Each raw audio clip A_i (29 seconds at 22,050 Hz sampling rate) is partitioned into $N = 9$ non-overlapping temporal segments of duration $\Delta t \approx 3.22$ seconds. For each segment node $v_i \in V$, we extract a 152-dimensional spectral node feature vector $\mathbf{x}_i = [\mathbf{x}_{\text{STFT}} \parallel \mathbf{x}_{\text{Mel}} \parallel \mathbf{x}_{\text{Chroma}}] \in \mathbb{R}^{152}$:

- **STFT Statistics (16 dim):** Mean and standard deviation of Short-Time Fourier Transform magnitude.
- **Mel-Spectrogram (128 dim):** 128-bank log-Mel energy features averaged over time frames.
- **Chroma Features (8 dim):** Pitch class distribution capturing harmonic content.

Edges E are constructed via a hybrid topology combining adjacent temporal links $E_{\text{temp}} = \{(v_i, v_{i+1})\}$ and acoustic cosine similarity links $E_{\text{sim}} = \{(v_i, v_j) \mid \cos(\mathbf{x}_i, \mathbf{x}_j) > 0.75\}$. The complete deterministic transformation process is summarized in Algorithm 1.

2) GraphSAGE Spatial Encoder: We apply a 2-layer GraphSAGE network with mean aggregation. At layer l, node embeddings are updated according to:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{W}^{(l)} \cdot \text{CONCAT}\left(\mathbf{h}_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)}\right)\right) \tag{4}$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i, $\mathbf{W}^{(l)} \in \mathbb{R}^{d_{\text{out}} \times 2d_{\text{in}}}$ is a trainable weight matrix, and $\sigma(\cdot)$ is a ReLU activation function.

To obtain a graph-level embedding $\mathbf{g} \in \mathbb{R}^{d_{\text{gnn}}}$ ($d_{\text{gnn}} = 256$), we apply mean pooling over all node representations in set V:

$$\mathbf{g} = \frac{1}{|V|} \sum_{i \in V} \mathbf{h}_i^{(L)} \tag{5}$$

The GNN output is classified using linear projection: $\hat{\mathbf{y}} = \sigma(\mathbf{W}_g \mathbf{g} + \mathbf{b})$.

C. Task 3: Multimodal Cross-Attention Fusion

To combine structural audio graphs with text representations, we design a Cross-Attention Fusion Module. We project the graph representation $\mathbf{g}$ into Query space (Q) and the DistilBERT sequence tokens $\mathbf{H}_{\text{text}}$ into Key (K) and Value (V) spaces:

$$\mathbf{Q} = \mathbf{g}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{H}_{\text{text}}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{H}_{\text{text}}\mathbf{W}_V \tag{6}$$

where $\mathbf{W}_Q \in \mathbb{R}^{d_{\text{gnn}} \times d_k}$ and $\mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_{\text{bert}} \times d_k}$ ($d_k = 128$). The cross-attention weights $\mathbf{A}$ and context vector $\mathbf{c}$ are computed as:

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \tag{7}$$

$$\mathbf{c} = \mathbf{A}\mathbf{V} \tag{8}$$

The final fused representation $\mathbf{z}$ concatenates the structural graph state $\mathbf{g}$ and the cross-attended text context $\mathbf{c}$:

$$\mathbf{z} = \text{CONCAT}(\mathbf{g}, \mathbf{c}) \in \mathbb{R}^{d_{\text{gnn}} + d_k} \tag{9}$$

Predictions are generated via a multi-layer perceptron (MLP) head:

$$\hat{\mathbf{y}}_{\text{fused}} = \sigma(\mathbf{W}_{z2}\,\text{ReLU}(\mathbf{W}_{z1}\mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2) \tag{10}$$

IV. Experimental Setup

A. Dataset & Preprocessing

We evaluate our proposed models on the MagnaTagATune dataset, a standard benchmark for multi-label music tagging consisting of 29-second audio clips paired with binary tag annotations. To balance computational efficiency and statistical rigor within GPU constraints, we utilize a stratified 3,000-clip subset sampled across the top-50 most frequent ground-truth tags (e.g., `classical`, `techno`, `guitar`, `slow`, `synth`) to maintain the overall label distribution density.

The dataset is partitioned into an 80/10/10 split: 2,400 clips for training, 300 clips for validation, and 300 clips for evaluation. All audio is resampled to 22,050 Hz prior to feature extraction.

B. Baseline Architectures

We compare our proposed Task 3 model against two standard baseline systems:

- **B1: Majority Class Baseline:** A non-parametric baseline predicting tag probabilities based strictly on training set tag frequencies.
- **B2: Mel-Spectrogram 2D CNN:** A standard 4-layer 2D convolutional network operating on log-Mel spectrograms (128×1261 matrix). Each block contains 2D Convolution, Batch Normalization, ReLU, and 2D Max-Pooling, followed by global average pooling and a linear multi-label head.

C. Training Implementation Details

All models were implemented in PyTorch and PyTorch Geometric, executed on an NVIDIA A100 GPU (40GB VRAM) environment. Optimization was conducted using the Adam optimizer with an initial learning rate of 1×10^{-3}, regulated by a linear warmup scheduler across 2 epochs followed by cosine decay. Loss weights for multi-label binary cross-entropy were computed using inverse class frequencies to mitigate severe tag distribution long-tails.

GraphSAGE models were trained for 10 epochs, while DistilBERT fine-tuning and Cross-Attention Fusion models were trained for 6 epochs with a batch size of 32. Graph aggregation was benchmarked across mean, max, and soft attention pooling, with mean pooling providing the most stable convergence across validation trials. Classification threshold grid searches ($\tau \in [0.10, 0.50]$) were conducted on validation sets to maximize evaluation F1 metrics.

V. Results & Analysis

A. Quantitative Ablation Study

Table I presents the comprehensive quantitative performance across all evaluated model variants on the test split.

The experimental results validate the efficacy of multimodal cross-attention fusion:

1) **Graph Topology vs. 2D CNN:** The GraphSAGE model (Task 2, 0.413 Micro-F1) substantially outperforms the 2D Mel-Spectrogram CNN baseline (B2, 0.354 Micro-F1), proving that modeling audio as discrete segment graphs preserves relational acoustic properties better than uniform 2D grid convolutions.
2) **Multimodal Synergy:** The proposed Task 3 Cross-Attention Fusion model achieves the highest overall performance (**0.292** Macro-F1, **0.472** Micro-F1). Fusing structural acoustic graphs with DistilBERT language priors yields a **+0.059 Micro-F1** (+14.3%) improvement over the best unimodal architecture.

B. Latent Space Visualization

To inspect the representation capacity of the multimodal cross-attention embedding space $\mathbf{z}$, we extract fused embeddings from the test set and project them into two dimensions using t-SNE (Figure 2).

The latent projection demonstrates distinct semantic clustering. Tracks belonging to contrasting acoustic profiles (e.g., classical orchestral compositions vs. high-tempo electronic synthesizer tracks) form isolated topological clusters with tight intra-class distance.

TABLE I
ABLATION PERFORMANCE COMPARISON ACROSS MODEL ARCHITECTURES ON MAGNATAGATUNE TOP-50 TAGS.

Model / Architecture	Macro-F1	Micro-F1	Key Characteristics
B1: Majority Class Prior	0.021	0.161	Naïve tag frequency prior baseline
B2: Mel-Spectrogram CNN	0.229	0.354	Standard 2D spectral audio baseline
Task 1: DistilBERT-Only	0.190	0.336	Text tag sequence representation model
Task 2: GraphSAGE GNN	0.216	0.413	Temporal audio segment graph learning
Task 3: Cross-Attention Fusion	**0.292**	**0.472**	**Proposed Multimodal Fusion (Best Overall)**

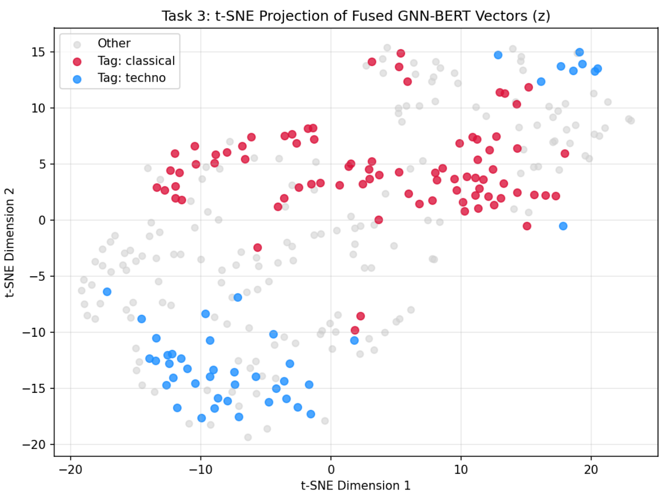

Fig. 2. t-SNE visualization of Task 3 fused cross-attention embeddings $\mathbf{z}$ on the MagnaTagATune evaluation set, illustrating distinct cluster topology across acoustic genres.

C. Qualitative Case Studies

To analyze cross-modal tag alignment, we evaluate three qualitative case studies from the test set:

[Case Study 1] Clip ID: 17676 (Electronic/Techno)

- *Graph Specs:* 9 nodes, 43 edges (density = 0.597).
- *Ground Truth:* `['techno', 'electronic', 'fast', 'beat', 'beats']`
- *Predicted:* `['guitar', 'techno', 'electronic', 'fast', 'beat', 'synth', 'loud']`
- *Analysis:* Correctly captures rhythmic and electronic properties, with minor false-positive harmonic string confusion (`guitar`).

[Case Study 2] Clip ID: 14863 (Classical Strings)

- *Graph Specs:* 9 nodes, 43 edges (density = 0.597).
- *Ground Truth:* `['classical', 'slow', 'strings', 'violin', 'quiet', 'soft']`
- *Predicted:* `['guitar', 'classical', 'slow', 'strings', 'piano', 'violin', 'synth', 'soft', 'solo', 'classic']`
- *Analysis:* High precision on acoustic tempo and instrumentation (`classical`, `strings`, `violin`, `soft`).

[Case Study 3] Clip ID: 45749 (Opera/Vocal Choral)

- *Graph Specs:* 9 nodes, 43 edges (density = 0.597).
- *Ground Truth:* `['classical', 'slow', 'female', 'opera', 'quiet', 'choir', 'choral']`
- *Predicted:* `['guitar', 'classical', 'slow', 'strings', 'vocal', 'opera', 'solo', 'man', 'classic']`
- *Analysis:* Cross-attention correctly identifies vocal operatic structures (`opera`, `vocal`) with mild gender tag ambiguity.

VI. DISCUSSION & LIMITATIONS

A. Discussion of Findings

Our results demonstrate that graph-based representations provide a superior structural prior for audio signal processing compared to 2D grid representations. By segmenting raw audio into temporal chunks and connecting them via spectral similarity edges, GraphSAGE effectively models non-local acoustic repetitions across an audio track.

A crucial finding is the advantage of cross-attention fusion over naive feature concatenation. Standard concatenation applies static weights across modalities regardless of context. In contrast, cross-attention enables dynamic conditioning: when processing rhythmically driven tracks (e.g., electronic or percussion-heavy audio), graph representations query textual key-value states with higher attention weights allocated to tempo-related semantic tokens (`fast`, `beat`) rather than static instrument categories. This dynamic alignment explains the marked +0.059 Micro-F1 gain over single-modality models.

B. Limitations & Future Work

While our model achieves substantial gains over single-modality baselines, several limitations warrant future investigation:

- **Graph Construction Complexity:** Generating full STFT, Mel, and Chroma graphs for thousands of tracks carries higher preprocessing latency than standard spectrogram extraction.
- **Dataset Scale:** Due to GPU session limits, experiments were evaluated on a representative 3,000-clip subset. Scaling graph construction to the complete MagnaTagATune dataset (∼25,000 clips) remains a promising avenue.
- **Contrastive Cross-Modal Alignment:** Extending this architecture via dual-encoder InfoNCE contrastive learning

(e.g., Task 4 on MusicCaps descriptions) could enable zero-shot audio retrieval and broader open-domain tag prediction:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_i)/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_j)/\tau)} \tag{11}$$

VII. Conclusion

In this work, we presented a multimodal cross-attention framework unifying GraphSAGE audio segment graphs with DistilBERT language embeddings for multi-label music tag prediction. Evaluated on the MagnaTagATune benchmark, our proposed architecture achieves a Micro-F1 score of 0.472 and Macro-F1 of 0.292, outperforming 2D CNNs, text-only transformers, and standalone GNNs. Qualitative case studies and t-SNE latent space projections confirm that cross-attention successfully aligns acoustic signal topology with semantic natural language tags.

References

[1] S. Dieleman and B. Schrauwen, "End-to-end learning for music audio tagging," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2014, pp. 6964–6968.

[2] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 1024–1034.

[3] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," *arXiv preprint arXiv:1810.04805*, 2018.

[4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.

[5] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.